

Turning Feature Attributions into Sufficient Explanations Using Conformal Prediction

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Abstract

Feature attribution methods are widely used for explaining image-based predictions, as they provide feature-level insights that can be intuitively visualized. However, such explanations often vary in their robustness and may fail to reflect the reasoning of the underlying black-box model faithfully. To address these limitations, we propose a novel conformal prediction-based approach that enables users to assign confidence levels to the important explanation regions. The proposed approach employs any feature-attribution method to identify a subset of salient features sufficient to preserve the model’s prediction, regardless of the information carried by the excluded features, without demanding access to ground-truth explanations for calibration. Four conformity functions are proposed to quantify the extent to which explanations conform to the model’s predictions. The approach is empirically evaluated using five explainers across six image datasets. The empirical results demonstrate that FastSHAP consistently outperforms the competing methods in terms of both fidelity and informational efficiency, the latter measured by the size of the explanation regions. Furthermore, the results reveal that conformity measures based on super-pixels are more effective than their pixel-wise counterparts.

Keywords: Conformal prediction, Explainable machine learning, Image classification.

1. Introduction

Many of the state-of-the-art machine learning algorithms operate as black boxes, which not only limits the user’s ability to understand the rationale behind their decisions but also constrains their adoption in high-stakes domains. The transparency is necessary not only for establishing trust in predictive models but also for addressing compliance with legal and regulatory requirements, as well as ethical obligations (Goodman and Flaxman, 2017). Therefore, explainable machine learning has emerged as a prominent research area that achieves interpretability while maintaining high predictive performance.

Explainable machine learning methods employ a set of strategies to make model behavior more understandable. Among the most common strategies are the construction of counterfactual examples (Karimi et al., 2020; Mothilal et al., 2020; Van Looveren and Klaise, 2021; Guo et al., 2021), the use of local interpretable surrogate models (Ribeiro et al., 2016; Lundberg and Lee, 2017), the identification of important features (Chen et al., 2018; Yoon et al., 2019; Jethani et al., 2021), and feature attribution methods (Simonyan et al., 2014; Sundararajan et al., 2017; Lundberg and Lee, 2017; Schwab and Karlen, 2019; Covert and Lee, 2021; Jethani et al., 2022).

Feature attribution methods are particularly prominent since they provide detailed information by assigning an importance score to each input feature, which can provide fine-grained insights into which specific inputs mainly drive the model’s decision. Moreover, the attributed importance scores lend themselves to intuitive visualizations, e.g., heatmaps for image data or bar plots for tabular inputs. However, feature attributions are not always robust (Hsieh et al., 2021; Fel et al., 2023), and their fidelity to the underlying black-box model may vary (Yeh et al., 2019; Lakkaraju and Bastani, 2020), i.e., the extent to which an explanation accurately captures the decision-making behavior of the underlying model. The low fidelity can have adverse consequences, especially in human-AI hybrid decision-making settings (Sadeghi et al., 2024; Cabitza et al., 2024). Generally, explanations are provided without associated uncertainty estimates, and methods that do offer post-hoc uncertainty quantification, e.g., (Alkhatib et al., 2023) and (Alkhatib et al., 2024), rely on access to ground-truth explanations for offline calibration, which may not always be feasible, particularly in the case of image data. Furthermore, such approaches preclude the user from controlling the fidelity of the generated explanations.

Therefore, the main contributions of this work are:

- **a novel method based on conformal prediction** that enables users to directly assign a confidence level to the important explanation regions while also providing an indication of their uncertainty
- **the method determines sufficient explanations** that can reproduce the predictions of the underlying black-box model at a user-specified confidence level
- **a set of conformity functions** specifically designed to quantify the degree to which the explanations conform to the model’s predictions

The next section introduces fundamental concepts of the conformal prediction framework and feature attribution explanations, along the way, we also introduce the notation used throughout the paper. Section 3 details the proposed method, while Section 4 outlines the experimental setup and presents the results of the empirical investigation. Section 5 provides a brief overview of related work. Finally, the concluding remarks summarize the main findings and discuss potential directions for future research.

2. Preliminaries

In this section, we provide a concise overview of the key concepts underlying the proposed approach. We begin by introducing conformal prediction in the following subsection, and subsequently present explanation methods based on feature attribution, with particular emphasis on the Shapley value as a representative example.

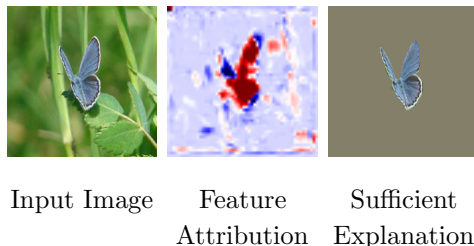


Figure 1: An example of the output explanation (subfigure c) using the proposed method with 85% confidence.

2.1. Conformal Prediction

Conformal prediction has emerged as a prominent approach for uncertainty estimation within the machine learning community. Originally introduced as a transductive approach, conformal prediction required training a separate model for each test instance, making it computationally expensive (Gammerman et al., 1998; Saunders et al., 1999). Consequently, the inductive conformal prediction has been proposed by Papadopoulos et al. (2002), which trains a single model on provided data and employs it for predictions on new instances. For simplicity, we refer to inductive conformal prediction as conformal prediction throughout the remainder of this manuscript. Conformal prediction methods construct prediction sets that include the true target with a predefined probability (a property referred to as validity), employing past observations to determine precise levels of confidence in new predictions (Shafer and Vovk, 2008). Under the assumption that a given calibration dataset $\mathbb{Z}$ consists of k independent and identically distributed (i.i.d.) pairs of inputs and labels $\mathbb{Z} = \{(x_1, y_1), (x_2, y_2), \dots, (x_k, y_k)\}$, a conformal predictor assigns a p-value to a candidate pair (x_{k+1}, y_{k+1}) , where y_{k+1} is a potential label for a new observation x_{k+1} drawn from the same distribution. The definition of p-value here is intertwined with the non-conformity measure (α_i), which quantifies the degree of 'strangeness' of an example, i.e., how unusual a data point appears relative to observed data examples (Tocaceli and Gammerman, 2017). There is no universal function for measuring non-conformity. Nevertheless, simple choices for the non-conformity measure are often sufficient. For instance, in regression problems, one may use the absolute error ($\alpha_i = |y_i - \tilde{y}_i|$) (Papadopoulos et al., 2002), where $\tilde{y}_i$ is the predicted outcome by the underlying model; or ($\alpha_i = 1 - P_f(y_i | x_i)$) in classification, where $P_f(y_i | x_i)$ denotes the assigned probability to label y_i by model f . Therefore, the p-value corresponding to a candidate label $y_{k+1} \in \mathbb{Y}$, where $\mathbb{Y}$ is the set of possible labels for each prediction, is defined by (Vovk, 2012):

$$p^{y_{k+1}} = \frac{|\{(x_i, y_i) \in \mathbb{Z} : \alpha_i \geq \alpha_{y_{k+1}}\}| + 1}{k + 1}.$$

Given a predefined significance level $\epsilon \in (0, 1)$ and a sequence of non-conformity scores $\mathbb{A} = \{\alpha_1, \alpha_2, \dots, \alpha_k\}$, the smallest $\alpha_\epsilon \in \mathbb{A}$ such that:

$$\frac{|\{(x_i, y_i) \in \mathbb{Z} : \alpha_i \leq \alpha_\epsilon\}| + 1}{k + 1} \geq 1 - \epsilon, \quad (1)$$

can be employed to construct a prediction set as follows: $\{y_{k+1} \in \mathbb{Y} : \alpha_{y_{k+1}} \leq \alpha_\epsilon\}$.

The specified non-conformity score (α_ϵ) in (1) guarantees that the resulting prediction sets contain the true label with a probability mass that meets or exceeds the specified confidence level $1 - \epsilon$.

Regardless of the non-conformity measure employed, the claim that the prediction sets contain the true label with confidence level $1 - \epsilon$ holds under the assumption that the data are i.i.d. (Shafer and Vovk, 2008). The claim also holds under the slightly relaxed assumption that the data samples are probabilistically exchangeable (Vovk et al., 2005). However, the choice of the non-conformity function impacts the efficiency of the conformal predictor (Linusson et al., 2020; Aleksandrova and Chertov, 2021).

2.2. Explanations via Feature Attribution

The proposed method can be applied to explanation techniques that produce feature attributions. Explaining model predictions through feature attribution is a class of explanation methods that assign a score to each input feature, reflecting the feature’s proportional contribution to the model’s output. A prominent example of feature attribution methods is the Shapley value, which can be used to construct an additive explanation model μ that serves as an interpretable approximation of a value function v . The value function quantifies how the model’s prediction changes when different subsets of features are marginalized out. The explanation model μ can be expressed as follows (Lundberg and Lee, 2017):

$$\mu(N) = \phi_0(v) + \sum_{i \in N} \phi_i(v),$$

where $\phi_0(v)$ is a constant and $\phi_i(v)$ the marginal contribution of feature $i \in N = \{1, 2, \dots, n\}$. $\mu(N)$ can be learned by minimizing the following weighted least squares loss function for a given instance $x \in X$ (Covert and Lee, 2021):

$$\mathcal{L}(v_x, \mu_x) = \sum_{S \subseteq N} \omega(S) \left(v_x(S) - \mu_x(S) \right)^2, \quad (2)$$

with ω a weighting kernel and $S \subseteq N$. Since the proposed approach involves selecting subsets of features, it is important to highlight standard approaches to remove features (an essential consideration in Shapley-value-based explanation methods). There are three commonly used value functions v for marginalizing out features that are not included in a coalition S :

1. Interventional approach (Marginal Expectations/Random Baseline) (Chen et al., 2020): $v_x(S) = \mathbb{E}_{x_S} [h(x_S, X_{N \setminus S}; \theta)]$, where h is a predictive model. This approach replaces the removed features with independent random values drawn from the marginal distribution, i.e., breaks dependence between selected and masked features, which may lead to unrealistic input combinations off the data manifold.
2. Observational approach (Conditional Expectations) (Chen et al., 2020; Zern et al., 2023): $v_x(S) = \mathbb{E}_{x_S} [h(X_S; \theta) | X_S = x_S]$. Here, the masked features are sampled conditionally based on the selected subset x_S , therefore, preserving dependencies present in the data, and the generated combinations remain in distribution.
3. Baseline removal approach (Sundararajan and Najmi, 2020): $v_x(S) = h(x_S, \tilde{x}_{N \setminus S}; \theta)$, where $\tilde{x}$ is a fixed baseline vector that often takes the values of $\mathbb{E}[X]$

We adopt the baseline removal approach due to its simplicity of implementation and compatibility with a broad range of feature attribution methods that do not rely on Shapley value estimation, e.g., CXPlain (Schwab and Karlen, 2019) and Integrated Gradients (Sundararajan et al., 2017), where the baseline can be conveniently set to a zero vector, particularly when standard scaling is applied and feature values are centered around zero.

The fidelity of the attributions to the underlying black-box model can vary both across different explanation methods and across individual data instances (Yeh et al., 2019; Hsieh

et al., 2021; Fel et al., 2023; Wang et al., 2024; Miró-Nicolau et al., 2024). Additionally, such attributions do not precisely determine the exact subset of features responsible for the prediction of the black-box model. The proposed method employs conformal prediction to identify the minimal subset of important input features, based on the attributed scores, that is sufficient to preserve the model’s prediction while ensuring that the explanation approximation error remains within a prespecified significance level.

3. The Proposed Method

In what follows, we adopt the notion of a conformity measure, which, while similar to the concept of non-conformity, serves a complementary purpose, i.e., rather than quantifying the strangeness of an example, the conformity measure quantifies how typical or common it appears in comparison to observed examples. *Our objective is to compute a conformity function that accurately identifies the smallest subset of features sufficient for the model to produce the same prediction as it would using all features for a given data point.* We begin with a trivial, theoretically motivated conformity function, and then build upon this foundation by introducing more advanced conformity functions. The proposed method operates on post-hoc explanations and grants users control over the acceptable error rate, defined as the discrepancy between the model’s original prediction and the prediction based on the identified important regions of the input features.

3.1. Algorithm

We define a function $\psi(x, \Phi; \sigma_\epsilon)$ that post-processes the explanations of a model h , where ψ is parameterized by σ_ϵ that controls the level of conservativeness, and Φ represents the set of attributed scores for the input features of x . Lower values of σ_ϵ lead to more conservative explanations, i.e., larger sets of features are identified as important to maintain the prediction. The output of ψ is a subset of features that contains the necessary features for model h to yield its prediction for a given test instance x , with the probability of omitting an essential feature upper bounded by ϵ .

The parameter σ_ϵ is learned using a held-out calibration set $\mathbb{Z} = \{(x_1, \Phi_1), (x_2, \Phi_2), \dots, (x_k, \Phi_k)\}$ over which we compute conformity measures $\sigma_i, \forall x_i \in \mathbb{Z}$. The conformity measures quantify how typical a feature is as part of a sufficient explanation relative to previously observed explanation examples. More specifically, for each data instance $x_i \in \mathbb{Z}$, the conformity function identifies the highest threshold σ_i such that the prediction made using only the subset S_i of features with all attribution scores $\phi_f^{(i)} \geq \sigma_i$ matches the prediction obtained when using the grand coalition of all features. Formally, let $\Phi = \{\phi_1, \phi_2, \dots, \phi_d\}$ denote the attribution scores assigned to the input features $x = \{f_1, f_2, \dots, f_d\}$, we define conformity measure σ_i for instance x_i as:

$$\sigma_i = \sup \left\{ \tau \in \Phi_i : S_{i,\tau} = \left\{ f_j \in x_i : \phi_j^{(i)} \geq \tau \right\}, \arg \max h(S_{i,\tau}) = \arg \max h(x_i) \right\}, \quad (3)$$

where τ can take values from quantile levels over the range of Φ_i values, or alternatively, from the discrete set of attribution scores in Φ_i . For a significance level $\epsilon \in (0, 1)$ and a set

of conformity scores $\sigma = \{\sigma_1, \sigma_2, \dots, \sigma_k\}$, we define σ_ϵ as follows:

$$\sigma_\epsilon = \sup \left\{ \sigma \in \sigma : \frac{|\{(x_i, \Phi_i) \in \mathbb{Z} : \sigma_i \geq \sigma\}| + 1}{k + 1} \geq 1 - \epsilon \right\}.$$

Finally, for new data examples x_{k+1} , ψ returns a sufficient explanation set $S_E^{(k+1)}$ that is constructed as follows:

$$S_E^{(k+1)} = \left\{ f_j \in x_{k+1} : \phi_j^{(k+1)} \geq \sigma_\epsilon \right\}.$$

Consequently, a feature j is included in the sufficient explanation coalition $S_E^{(k+1)}$ if its attribution score satisfies $\phi_j \geq \sigma_\epsilon$, in which case its corresponding value f_j is retained. Otherwise, the feature is omitted from $S_E^{(k+1)}$, and its value remains at its baseline value in $\tilde{x}$. The explanation returned by ψ is guaranteed to include the important features, sufficient to maintain the prediction, with probability at least $1 - \epsilon$. Nevertheless, a certain explainer tends to produce concise and informative explanation sets $S_E^{(k+1)}$, whereas an explainer that assigns importance scores at random may result in excessively large sets, potentially encompassing the entire input. An obvious limitation of explanations based on individual important pixels is that they often result in sparse and, possibly, noisy representations both for the user and the predictive model. Consequently, we also propose the subsequent conformity measures.

3.2. Super-Pixel-Based Conformity Function

While the default conformity function used in the proposed algorithm (as defined in (3)) determines the explanation set $S_E^{(k+1)}$, it may result in highly fragmented image regions. Therefore, we introduce an alternative conformity function that incorporates the image's segmentation structure, thereby enabling ψ to return a set of superpixels as a coherent and sufficient explanation.

The super-pixel-based conformity function applies any standard image segmentation algorithm to compute superpixels $\gamma_l \subset x_i$ and defines $\sigma_i^{(sp)}$ as the largest attribution threshold such that the model's prediction, restricted to the subset S_i of superpixels that contain at least a number ρ of features satisfying $\phi_j^{(i)} \geq \sigma_i^{(sp)}$ (i.e., $|\{\phi_j^{(i)} \in \gamma_l : \phi_j^{(i)} \geq \sigma_i^{(sp)}\}| \geq \rho$), matches the prediction obtained using the grand coalition of all features. $\sigma_i^{(sp)}$ is defined as follows:

$$\sigma_i^{(sp)} = \sup \left\{ \tau \in \Phi_i : S_{i,\tau} = \left\{ \gamma_l \subset x_i : |\{\phi_j^{(i)} \in \gamma_l : \phi_j^{(i)} \geq \tau\}| \geq \rho \right\}, \arg \max h(S_{i,\tau}) = \arg \max h(x_i) \right\}, \quad (4)$$

where τ as mentioned before, can take values from the attribution scores within Φ_i or can be the quantile levels over the range of Φ_i , and the sufficient explanation set $S_E^{(k+1)}$ for a new data example x_{k+1} is formed as follows:

$$S_E^{(k+1)} = \left\{ \gamma_l \subset x_{k+1} : |\{\phi_j^{(k+1)} \in \gamma_l : \phi_j^{(k+1)} \geq \sigma_\epsilon^{(sp)}\}| \geq \rho \right\}.$$

3.3. Scaled Attribution Scores (Scaled Values)

This function operates similarly to (4), but the attribution scores of each data instance are standardized by centering them to a zero mean and scaling them to unit variance. Each explanation Φ_i is normalized by subtracting its mean ($\bar{\varphi}_i$) and dividing by its standard deviation (std_i) as $\Phi_i^{scaled} = \frac{\Phi_i - \bar{\varphi}_i}{std_i}$.

3.4. Summation Based Function (Summed Values)

Instead of returning a threshold for each data instance, the summation-based conformity function (σ_i^Σ) identifies the smallest subset of features, ranked by their attribution scores, whose cumulative sum is minimal and sufficient to preserve the model’s original prediction. σ_i^Σ is formally defined in (5).

$$\sigma_i^\Sigma = \inf \left\{ \sum_{f_j \in S_{i,\tau}} |\phi_j^{(i)}| : S_{i,\tau} = \{f_j \in x_i : \phi_j^{(i)} \geq \tau\}, \arg \max h(S_{i,\tau}) = \arg \max h(x_i) \right\}. \quad (5)$$

Then σ_ϵ^Σ is defined as follows: $\inf \left\{ \sigma^\Sigma \in \sigma : \frac{|\{(x_i, \Phi_i) \in \mathbb{Z} : \sigma_i^\Sigma \leq \sigma^\Sigma\}| + 1}{k+1} \geq 1 - \epsilon \right\}$, and for a sufficient explanation set $S_E^{(k+1)}$ for a new data instance x_{k+1} let $\pi_{k+1} = [f_1, f_2, \dots, f_d]$ be the list of features ordered such that $\phi_1 \geq \phi_2 \geq \dots \geq \phi_d$. Then, the sufficient explanation set is: $S_E^{(k+1)} = \left\{ f_j \in \pi_{k+1} : \left(\sum_{r=1}^j \phi_r \right) \leq \sigma_\epsilon^\Sigma \right\}$.

4. Empirical Investigation

We assess the proposed approach using 4 conformity functions across 6 distinct image datasets and 5 different explainers. The evaluation compares both the conformity functions and the explainers in terms of their informativeness, where producing more concise explanation sets corresponds to higher informativeness, and the fidelity of the determined explanations, i.e., the degree to which the explanations preserve the same prediction of the model when provided with the full input image.

4.1. Experimental Setup

Each dataset is partitioned into training, validation, and evaluation subsets. The training set is used to fit the predictive black-box models, the validation set serves to monitor overfitting and determine early stopping, and the evaluation set is reserved for assessing model performance. The evaluation set is further divided into a calibration subset and a test subset, where the calibration subset is employed to compute the conformity score at a predefined significance level, while the test subset is used to evaluate both the proposed conformity functions and the employed explainers. For the feasibility of the experiments, τ in Equations (3), (4), and (5) takes values from the quantile levels over the range of Φ_i . Superpixel segmentation is performed using the SLIC algorithm (Achanta et al., 2012). All images are normalized with mean and standard deviation of the training set. The black-box models are based on ResNet50 (He et al., 2016). The employed explainers are

Saliency (Simonyan et al., 2014), InputXGradient (Shrikumar et al., 2016), KernelSHAP, GradientSHAP (Lundberg and Lee, 2017), and FastSHAP (Jethani et al., 2022). A detailed description of the employed datasets is provided in Appendix E.

The results are evaluated based on the informativeness of the resulting explanation sets, adopting a paradigm analogous to the evaluation of the informational efficiency of conformal predictors, where efficiency is measured by the size of the prediction regions and sets, i.e., smaller prediction regions are more informative (Linusson et al., 2020; Messoudi et al., 2020; Alkhatib et al., 2023). For example, Figure 2 presents the explanations generated by 2 competing explainers for a prediction of an English Springer image taken from the Imagenette dataset. Adjacent to each explanation, we display its corresponding extracted S_E region obtained using our proposed approach. The extracted S_E regions in subfigures (c) and (e) successfully reproduce the original prediction of the English Springer. However, the S_E region derived using FastSHAP in subfigure (c) is smaller, thereby demonstrating superior informational efficiency. In the following experiments, the fidelity quantifies the empirical agreement between the model’s prediction on the sufficient explanation (S_E) and the full input. Formally, fidelity is computed as the ratio between the number of test instances for which both predictions coincide and the total number of test instances.

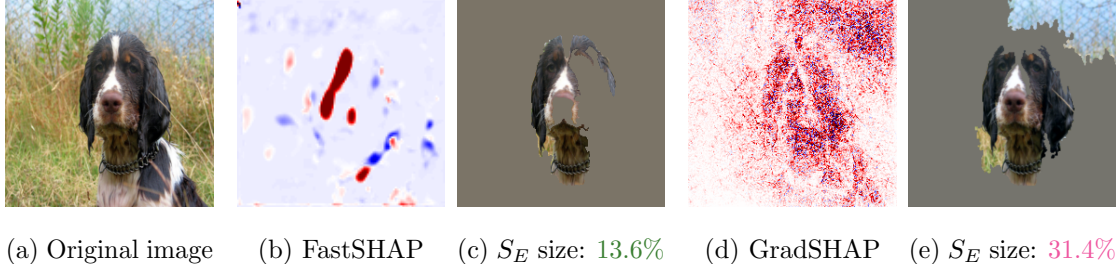


Figure 2: Explanations generated by FastSHAP and GradientSHAP for an English Springer image from the Imagenette dataset, along with their corresponding extracted S_E regions at 95% confidence obtained using the Scaled Values conformity function. The shown S_E regions successfully reproduce the original prediction as an English Springer.

4.2. Informational Efficiency Evaluation

We evaluate the informational efficiency of the conformal explainers using several baseline approaches (KernelSHAP, FastSHAP, GradientSHAP, Saliency, and InputXGradient), while also evaluating the efficiency of the proposed conformity measures in decisively distinguishing between conforming and non-conforming patterns. Figure 3 shows the average size of the extracted explanation regions (S_E) at different confidence levels for each explainer using the Scaled Values function. The detailed results using the four conformity functions at 95% confidence level are available in Table 1 in Appendix A. The results indicate that FastSHAP is generally a more efficient explainer than the competing approaches, while KernelSHAP shows high uncertainty and produces substantially larger S_E regions, therefore, its explanations are comparatively less informative.

The comparison of the proposed conformity functions, as illustrated in Figure 4, indicates that the super-pixel-based functions (Super-Pixels and Scaled Values) result in more

informationally efficient explanations than the pixel-based functions (Pixelwise and Summed Values) at high confidence levels. Particularly, by aggregating information over coherent regions rather than evaluating individual pixels in isolation, the super-pixel-based measures can capture higher-level structural patterns, which leads to explanations that are more compact and less noisy. In contrast, pixel-based measures tend to produce fragmented and less efficient explanations, as the importance values can be distributed across many fine-grained elements, as illustrated in Figure 5. Moreover, applying scaling to the obtained attribution values (via the Scaled Values function) serves to regulate their magnitudes, thereby reducing the size of the determined explanation regions at higher confidence levels, as detailed in Table 1 and Table 3.

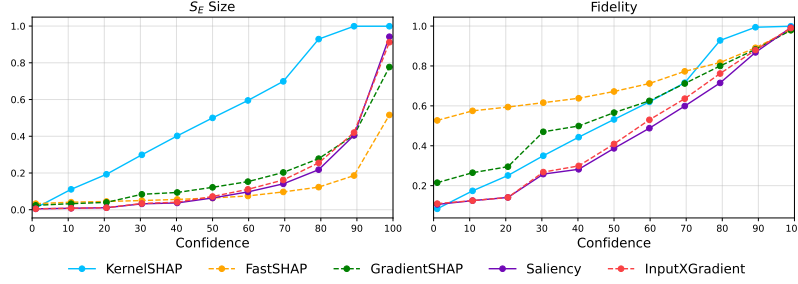


Figure 3: The effect of varying confidence levels on the size of S_E and the corresponding fidelity levels, obtained using the Scaled Values conformity function on the Animal-10 dataset.

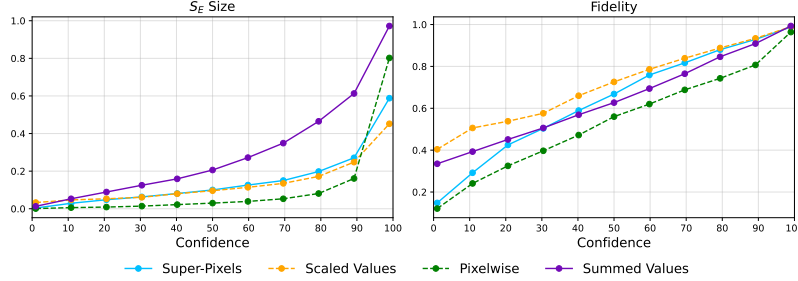


Figure 4: The effect of the confidence level on the size of S_E and the fidelity level of the retrieved explanations using FastSHAP with the proposed conformity functions on Imagenette dataset.

4.3. Fidelity Evaluation

The confidence levels allow the user to explicitly control the trade-off between compactness and reliability of the explanations. As shown in Figure 3, FastSHAP achieves high fidelity levels while simultaneously maintaining relatively small explanation regions. Additionally, Figure 4 illustrates that the Scaled Values function offers comparatively higher fidelity levels while maintaining relatively small S_E sizes. In contrast, the Summed Values function is the least efficient and results in the largest S_E sizes.

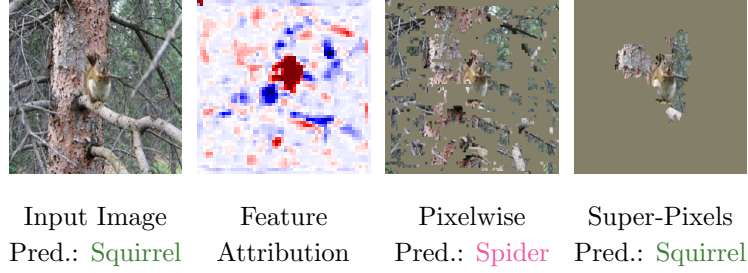


Figure 5: The sufficient explanation (S_E) determined using the pixel-wise conformity function (subfigure c) and the super-pixel-based function with scaled values (subfigure d). The example is obtained from the Animal-10 dataset at 95% confidence.

The comparison of explainers in terms of fidelity, as shown in Table 2, reveals comparable accuracy in reproducing the black-box model’s predictions across the same conformity function. However, super-pixel-based functions produce fidelity levels that are more consistent with the predefined confidence level than pixel-wise functions. Therefore, the super-pixel-based functions are shown not only to be more decisive in separating conforming from non-conforming patterns but also more reliable in maintaining the validity guarantees. Table 2 presents the detailed fidelity evaluation results for the compared explainers across the four proposed conformity functions, and Table 4 presents the fidelity results of FastSHAP at different confidence levels. The results are also summarized for all the datasets and the conformity functions in Figure 7. Figure 6 provides an illustrative example from the Imagenette dataset, demonstrating how increasing the confidence level, and thereby enforcing stricter fidelity guarantees, affects the resulting explanations.

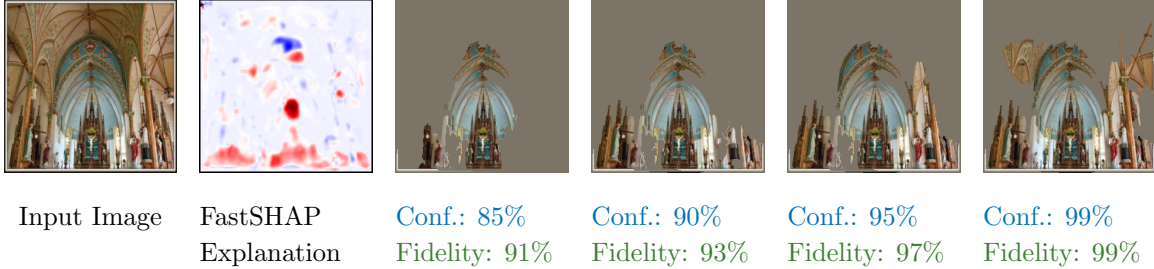


Figure 6: The effect of varying confidence levels on the trade-off between the fidelity and compactness of explanations. Fidelity is reported on the test set for each confidence level. The example is obtained from the Imagenette dataset.

5. Related Work

The conformal prediction framework has increasingly been employed as a means to quantify the uncertainty associated with explanations, as it provides distribution-free and finite-sample valid guarantees. Alkhatib et al. (2023) and Idrissi et al. (2025) combined the conformal prediction framework with feature attributions to derive feature-wise uncertainty estimates, thereby quantifying the reliability of individual attribution scores. In contrast,

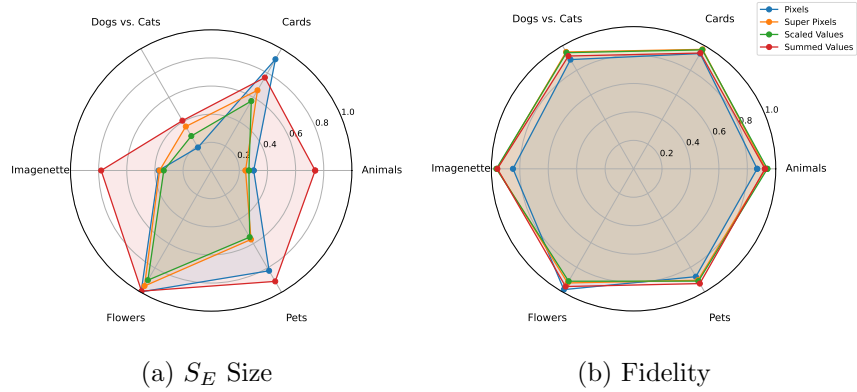


Figure 7: The average size and fidelity of the explanation regions determined at 95% confidence using FastSHAP. In subfigure (a), the Scaled Values function generally produces the most efficient, whereas in subfigure (b), all conformity functions achieve comparable levels of fidelity.

Alkhatib et al. (2024) employed conformal prediction to generate uncertainty indicators at the level of the entire explanation encompassing all the attributed scores. The Venn prediction framework (Vovk et al., 2003; Lambrou et al., 2015), a statistical framework closely related to conformal prediction for quantifying uncertainty, has also been explored for calibrating feature attributions (Löfström et al., 2024; Löfström et al., 2024). The impact of employing conformal prediction and Venn prediction to calibrate the predictions of the underlying black-box model, and consequently the generated explanations, has been investigated by Löfström et al. (2023) and Mehdiyev et al. (2025a). Moreover, the conformal prediction framework has been applied to provide fidelity guarantees for explanatory rules extracted from regression models (Johansson et al., 2022), while Alkhatib et al. (2022) proposed quantifying the uncertainty of post-hoc explanatory rules using the Venn prediction framework. Nevertheless, the mentioned approaches have been developed primarily for tabular data, and their applicability to image datasets remains limited.

Beyond the use of the conformal prediction framework for uncertainty quantification of explanations, research at the intersection of interpretable machine learning and conformal prediction has explored a variety of approaches. Prior work has either combined the predictions of interpretable models with conformal prediction or employed conformal prediction to enhance the interpretability of the underlying model (Johansson et al., 2019a,b; Martinez Gil et al., 2024; Narteni et al., 2026). In addition, Jaramillo and Smirnov (2021) proposed employing Shapley values as conformity scores for inductive conformal predictors, highlighting the potential of integrating attribution methods with conformal prediction.

Apart from conformal prediction, alternative approaches for quantifying the uncertainty of the explanations have been proposed. Schulz et al. (2022) investigated quantifying the uncertainties of surrogate model explanations. Hill et al. (2024) proposed to integrate uncertainty due to the complexity of the decision boundary with the uncertainty that arises from the approximation of the explanation function. For image classification, Zhang et al. (2022) developed an uncertainty quantification-based framework to interpret deep neural network decisions. Additionally, Mehdiyev et al. (2025b) combined predictive uncertainty with post-hoc explanations to quantify and explain the uncertainty in the predictions of a

machine learning model. However, such approaches are typically not model-agnostic, lack the statistical guarantees offered by the conformal prediction framework, and do not enable users to explicitly assign a confidence level to the generated explanations.

6. Concluding Remarks

We have introduced a novel algorithm based on conformal prediction that determines explanation regions sufficient to preserve the predictions of the underlying black-box model. The proposed algorithm enables users to balance fidelity and informativeness in the resulting explanations. Moreover, it does not require ground-truth annotations for calibration. To this end, we introduce four conformity functions designed to quantify the extent to which the attributed importance scores align with the model’s predictions. Our empirical evaluation reveals that conformity functions based on super-pixels consistently outperform pixel-based functions in terms of both efficiency and fidelity. Additionally, the results indicate that FastSHAP outperforms the competing explainers with respect to informational efficiency. Future research directions include the development of explanation algorithms that explicitly incorporate uncertainty information, thereby enabling the construction of more robust and reliable explanations. Another promising direction is the training of real-time explainers, e.g., FastSHAP, using super-pixels derived.

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Appendix A. Informational Efficiency Evaluation

The study evaluates the efficiency of conformal explainers with (KernelSHAP, FastSHAP, GradientSHAP, Saliency, and InputXGradient) using the proposed conformity measures. Results show that FastSHAP generally produces the most efficient explanations. Among conformity functions, super-pixel-based measures produce more compact explanations compared to pixel-based measures, which are fragmented and less efficient. Additionally, scaling attribution values further reduces explanation size by regulating their magnitude.

Table 1: The efficiency of the five explainers at 95% confidence level.

Conformity Function	Dataset	KernelSHAP	FastSHAP	GradientSHAP	Saliency	InputXGradient
Super-Pixels	Animals 10	0.945 $\pm$ 0.228	0.244 $\pm$ 0.1	0.675 $\pm$ 0.246	0.687 $\pm$ 0.241	0.659 $\pm$ 0.238
	Cards	0.989 $\pm$ 0.106	0.658 $\pm$ 0.126	0.935 $\pm$ 0.099	0.932 $\pm$ 0.096	0.938 $\pm$ 0.094
	Dogs and Cats	0.955 $\pm$ 0.207	0.361 $\pm$ 0.134	0.49 $\pm$ 0.244	0.502 $\pm$ 0.268	0.461 $\pm$ 0.241
	Imagenette	0.955 $\pm$ 0.208	0.367 $\pm$ 0.181	0.641 $\pm$ 0.257	0.589 $\pm$ 0.264	0.561 $\pm$ 0.257
	Flowers 102	0.957 $\pm$ 0.202	0.948 $\pm$ 0.122	0.822 $\pm$ 0.187	0.824 $\pm$ 0.184	0.79 $\pm$ 0.178
	Oxford IIIT Pet	0.959 $\pm$ 0.197	0.566 $\pm$ 0.146	0.681 $\pm$ 0.246	0.67 $\pm$ 0.255	0.673 $\pm$ 0.231
Scaled Values	Animals 10	0.944 $\pm$ 0.23	0.268 $\pm$ 0.123	0.592 $\pm$ 0.149	0.583 $\pm$ 0.14	0.583 $\pm$ 0.152
	Cards	0.928 $\pm$ 0.258	0.57 $\pm$ 0.155	0.904 $\pm$ 0.099	0.953 $\pm$ 0.059	0.955 $\pm$ 0.067
	Dogs and Cats	0.957 $\pm$ 0.202	0.283 $\pm$ 0.11	0.308 $\pm$ 0.093	0.305 $\pm$ 0.09	0.33 $\pm$ 0.101
	Imagenette	0.95 $\pm$ 0.219	0.337 $\pm$ 0.168	0.384 $\pm$ 0.125	0.479 $\pm$ 0.145	0.499 $\pm$ 0.156
	Flowers 102	0.957 $\pm$ 0.203	0.9 $\pm$ 0.189	0.873 $\pm$ 0.105	0.837 $\pm$ 0.098	0.833 $\pm$ 0.116
	Oxford IIIT Pet	0.953 $\pm$ 0.211	0.547 $\pm$ 0.145	0.678 $\pm$ 0.148	0.689 $\pm$ 0.146	0.724 $\pm$ 0.148
Pixelwise	Animals 10	0.948 $\pm$ 0.222	0.302 $\pm$ 0.079	0.953 $\pm$ 0.019	0.824 $\pm$ 0.088	0.957 $\pm$ 0.02
	Cards	0.992 $\pm$ 0.087	0.913 $\pm$ 0.029	0.97 $\pm$ 0.014	0.858 $\pm$ 0.115	0.977 $\pm$ 0.01
	Dogs and Cats	0.957 $\pm$ 0.204	0.189 $\pm$ 0.079	0.884 $\pm$ 0.042	0.793 $\pm$ 0.085	0.899 $\pm$ 0.044
	Imagenette	0.958 $\pm$ 0.202	0.374 $\pm$ 0.148	0.795 $\pm$ 0.082	0.674 $\pm$ 0.08	0.81 $\pm$ 0.09
	Flowers 102	0.977 $\pm$ 0.149	0.996 $\pm$ 0.006	0.992 $\pm$ 0.011	0.98 $\pm$ 0.021	0.99 $\pm$ 0.01
	Oxford IIIT Pet	0.989 $\pm$ 0.104	0.823 $\pm$ 0.083	0.982 $\pm$ 0.019	0.821 $\pm$ 0.146	0.981 $\pm$ 0.021
Summed Values	Animals 10	0.984 $\pm$ 0.031	0.739 $\pm$ 0.39	0.949 $\pm$ 0.162	0.926 $\pm$ 0.199	0.942 $\pm$ 0.177
	Cards	0.97 $\pm$ 0.056	0.763 $\pm$ 0.361	0.979 $\pm$ 0.091	0.913 $\pm$ 0.224	0.979 $\pm$ 0.088
	Dogs and Cats	0.996 $\pm$ 0.004	0.409 $\pm$ 0.419	0.945 $\pm$ 0.172	0.946 $\pm$ 0.151	0.95 $\pm$ 0.166
	Imagenette	0.975 $\pm$ 0.043	0.782 $\pm$ 0.384	0.865 $\pm$ 0.273	0.877 $\pm$ 0.249	0.907 $\pm$ 0.235
	Flowers 102	0.993 $\pm$ 0.002	0.99 $\pm$ 0.0	0.99 $\pm$ 0.0	0.99 $\pm$ 0.0	0.99 $\pm$ 0.0
	Oxford IIIT Pet	0.991 $\pm$ 0.016	0.91 $\pm$ 0.244	0.99 $\pm$ 0.0	0.945 $\pm$ 0.156	0.99 $\pm$ 0.0

Appendix B. The Fidelity of the Determined Explanations

The fidelity comparison shows that while all explainers achieve comparable accuracy in reproducing model predictions, super-pixel-based conformity functions provide more consistent fidelity with the confidence level than pixel-wise functions. Therefore, the super-pixel-based conformity functions are more decisive in distinguishing conforming vs. non-conforming explanations and more reliable in preserving validity guarantees than the pixel-wise functions.

Table 2: The fidelity of the explanations obtained using the five methods at 95% confidence level.

Conformity	Dataset	KernelSHAP	FastSHAP	Grad.SHAP	Saliency	InputXGrad.
Super-Pixels	Animals 10	0.957	0.93	0.954	0.956	0.946
	Cards	0.989	0.985	0.97	0.936	0.857
	Dogs and Cats	1	0.949	0.95	0.963	0.947
	Imagenette	0.957	0.96	0.97	0.961	0.96
	Flowers 102	0.958	0.927	0.914	0.917	0.913
	Oxford IIIT Pet	0.964	0.905	0.924	0.939	0.937
Scaled Values	Animals 10	0.948	0.941	0.946	0.936	0.932
	Cards	0.928	0.977	0.966	0.909	0.845
	Dogs and Cats	0.958	0.943	0.901	0.913	0.902
	Imagenette	0.955	0.965	0.94	0.946	0.938
	Flowers 102	0.957	0.913	0.914	0.9	0.902
	Oxford IIIT Pet	0.955	0.912	0.91	0.918	0.921
Pixelwise	Animals 10	0.958	0.869	0.905	0.899	0.907
	Cards	0.989	0.989	0.936	0.811	0.736
	Dogs and Cats	1	0.887	0.787	0.898	0.806
	Imagenette	0.96	0.848	0.902	0.872	0.904
	Flowers 102	0.978	0.981	0.811	0.845	0.788
	Oxford IIIT Pet	0.99	0.876	0.846	0.88	0.848
Summed Values	Animals 10	0.973	0.924	0.932	0.926	0.924
	Cards	0.992	0.974	0.94	0.902	0.838
	Dogs and Cats	0.987	0.916	0.915	0.936	0.917
	Imagenette	0.989	0.958	0.908	0.916	0.93
	Flowers 102	0.794	0.957	0.757	0.858	0.732
	Oxford IIIT Pet	0.93	0.932	0.884	0.911	0.889

Appendix C. The Effect of Confidence Level on the Efficiency

Changing the confidence level enables balancing explanation conciseness against reliability, as higher confidence levels ensure stronger fidelity guarantees, while lower levels yield more concise but less faithful explanations. The results in [Table 3](#) indicate that super-pixel-based functions consistently align better with the specified confidence levels than pixel-based ones.

Table 3: The efficiency of FastSHAP explanations at different confidence levels.

Conformity Function	Dataset	99%	95%	90%	85%
Super-Pixels	Animals 10	0.49 $\pm$ 0.165	0.244 $\pm$ 0.1	0.175 $\pm$ 0.075	0.142 $\pm$ 0.062
	Cards	0.899 $\pm$ 0.08	0.658 $\pm$ 0.126	0.486 $\pm$ 0.135	0.382 $\pm$ 0.127
	Dogs and Cats	0.735 $\pm$ 0.139	0.361 $\pm$ 0.134	0.254 $\pm$ 0.117	0.2 $\pm$ 0.104
	Imagenette	0.588 $\pm$ 0.187	0.367 $\pm$ 0.181	0.271 $\pm$ 0.157	0.226 $\pm$ 0.141
	Flowers 102	0.988 $\pm$ 0.047	0.948 $\pm$ 0.122	0.865 $\pm$ 0.199	0.775 $\pm$ 0.25
	Oxford IIIT Pet	0.882 $\pm$ 0.113	0.566 $\pm$ 0.146	0.45 $\pm$ 0.136	0.374 $\pm$ 0.128
Scaled Values	Animals 10	0.516 $\pm$ 0.18	0.268 $\pm$ 0.123	0.186 $\pm$ 0.089	0.149 $\pm$ 0.071
	Cards	0.855 $\pm$ 0.088	0.57 $\pm$ 0.155	0.415 $\pm$ 0.156	0.33 $\pm$ 0.136
	Dogs and Cats	0.447 $\pm$ 0.141	0.283 $\pm$ 0.11	0.216 $\pm$ 0.099	0.172 $\pm$ 0.089
	Imagenette	0.452 $\pm$ 0.192	0.337 $\pm$ 0.168	0.248 $\pm$ 0.136	0.202 $\pm$ 0.117
	Flowers 102	0.99 $\pm$ 0.053	0.9 $\pm$ 0.189	0.819 $\pm$ 0.257	0.762 $\pm$ 0.289
	Oxford IIIT Pet	0.82 $\pm$ 0.128	0.547 $\pm$ 0.145	0.434 $\pm$ 0.13	0.373 $\pm$ 0.119
Pixelwise	Animals 10	0.834 $\pm$ 0.055	0.302 $\pm$ 0.079	0.166 $\pm$ 0.051	0.118 $\pm$ 0.038
	Cards	0.99 $\pm$ 0.006	0.913 $\pm$ 0.029	0.642 $\pm$ 0.069	0.416 $\pm$ 0.05
	Dogs and Cats	0.467 $\pm$ 0.14	0.189 $\pm$ 0.079	0.114 $\pm$ 0.058	0.083 $\pm$ 0.047
	Imagenette	0.802 $\pm$ 0.171	0.374 $\pm$ 0.148	0.161 $\pm$ 0.072	0.108 $\pm$ 0.055
	Flowers 102	0.999 $\pm$ 0.002	0.996 $\pm$ 0.006	0.993 $\pm$ 0.009	0.99 $\pm$ 0.013
	Oxford IIIT Pet	0.978 $\pm$ 0.015	0.823 $\pm$ 0.083	0.662 $\pm$ 0.145	0.528 $\pm$ 0.157
Summed Values	Animals 10	0.947 $\pm$ 0.182	0.739 $\pm$ 0.39	0.586 $\pm$ 0.436	0.454 $\pm$ 0.436
	Cards	0.924 $\pm$ 0.219	0.763 $\pm$ 0.361	0.633 $\pm$ 0.408	0.501 $\pm$ 0.41
	Dogs and Cats	0.6 $\pm$ 0.419	0.409 $\pm$ 0.419	0.315 $\pm$ 0.393	0.257 $\pm$ 0.366
	Imagenette	0.972 $\pm$ 0.127	0.782 $\pm$ 0.384	0.613 $\pm$ 0.456	0.529 $\pm$ 0.467
	Flowers 102	0.99 $\pm$ 0.0	0.99 $\pm$ 0.0	0.99 $\pm$ 0.0	0.99 $\pm$ 0.0
	Oxford IIIT Pet	0.969 $\pm$ 0.131	0.91 $\pm$ 0.244	0.854 $\pm$ 0.308	0.784 $\pm$ 0.363

Appendix D. The Effect of Confidence Level on the Fidelity

The super-pixel-based functions yield fidelity levels that align more closely with the predefined confidence levels than the pixel-wise function.

Table 4: The fidelity of FastSHAP explanations at different confidence levels.

Conformity Function	Dataset	99%	95%	90%	85%
Super-Pixels	Animals 10	0.974	0.93	0.882	0.844
	Cards	0.985	0.97	0.936	0.857
	Dogs and Cats	0.981	0.949	0.92	0.903
	Imagenette	0.988	0.96	0.929	0.901
	Flowers 102	0.98	0.927	0.861	0.8
	Oxford IIIT Pet	0.969	0.905	0.863	0.814
Scaled Values	Animals 10	0.98	0.941	0.891	0.854
	Cards	0.977	0.966	0.909	0.845
	Dogs and Cats	0.976	0.943	0.914	0.891
	Imagenette	0.989	0.965	0.934	0.91
	Flowers 102	0.981	0.913	0.853	0.817
	Oxford IIIT Pet	0.964	0.912	0.866	0.834
Pixelwise	Animals 10	0.952	0.869	0.798	0.743
	Cards	0.989	0.936	0.811	0.736
	Dogs and Cats	0.956	0.887	0.841	0.812
	Imagenette	0.964	0.848	0.807	0.777
	Flowers 102	0.995	0.981	0.974	0.968
	Oxford IIIT Pet	0.968	0.876	0.777	0.705
Summed Values	Animals 10	0.98	0.924	0.872	0.814
	Cards	0.974	0.94	0.902	0.838
	Dogs and Cats	0.962	0.916	0.876	0.839
	Imagenette	0.993	0.958	0.909	0.876
	Flowers 102	0.956	0.956	0.956	0.956
	Oxford IIIT Pet	0.97	0.932	0.894	0.842

Appendix E. Dataset Details and Hardware Specifications

The experiments were conducted in a Python environment on a system equipped with an Intel® Xeon® Gold 6226R CPU @ 2.90GHz, with GPU support from an NVIDIA® Tesla V100. Table 5 presents an overview of the datasets used in the experiments.

Table 5: The dataset information.

Dataset	# Classes	Dataset Size	Train. Set	Val. Set	Test Set	Cal. Set	URL
Animals 10	10	26,179	19,895	1,048	3,900	1,336	Link
Cards	53	8,154	6,099	265	265	1,525	Link
Dogs and Cats	2	37,500	23,500	1,500	9,312	3,188	Link
Imagenette	10	13,394	8,522	947	1,001	2,924	Link
Flowers 102	102	8,189	1,020	1,020	4,581	1,568	Link
Oxford IIIT Pet	37	7,349	3,312	368	2,733	936	Link

Appendix F. Plots for the Effect of Confidence Level on the Efficiency

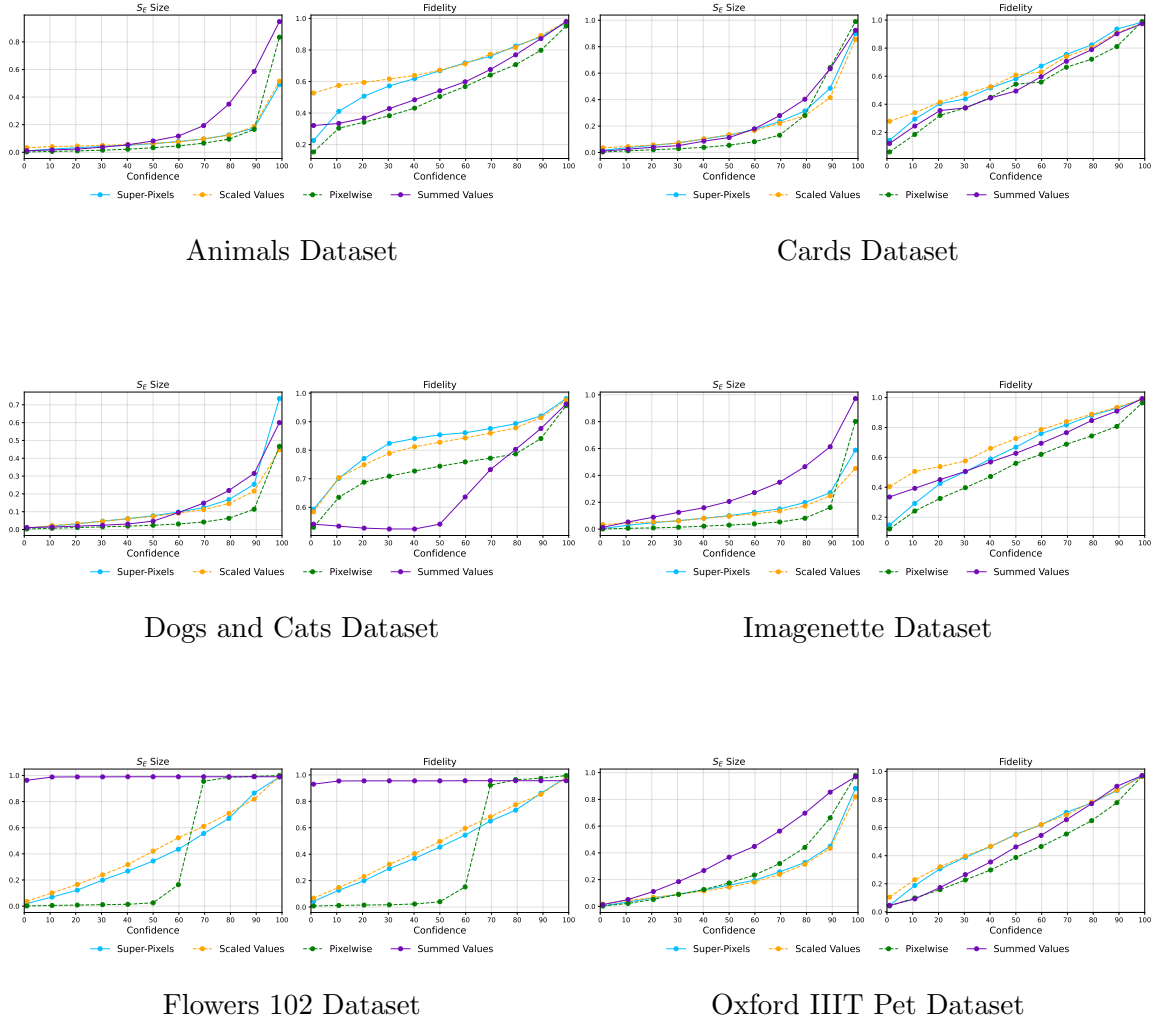


Figure 8: The effect of the confidence level on the size of S_E and the fidelity level of the retrieved explanations using FastSHAP with the four proposed conformity functions.